e-ISSN: 2582-5208
International Research Journal of Modernization in Engineering Technology and Science
(Peer-Reviewed, Open Access, Fully Refereed International Journal)
Volume:08/Issue:03/March-2026 Impact Factor- 8.187 www.irjmets.com

MergedNet: Multi-Backbone Feature Fusion and Repository Data Integrity Audit for Oral Image Classification

Rakshith Y B[1], Lohith R C[2], Keerthi A[3], Aishwarya S[4]

[1,2,3] Students , Department of Computer Science & Engineering, Kalpataru Institute of Technology Tiptur-572201,Karnataka ,India

[4] Assistant Professor, Department of Computer Science & Engineering, Kalpataru Institute of Technology Tiptur-572201, Karnataka, India

Affiliated with Visvesvaraya Technological University (VTU), Belagavi

ABSTRACT

Oral cancer detection using deep learning holds promise for assisting frontline health personnel, yet academic reporting requires strict data integrity and architectural transparency. We propose MergedNet, a multi-backbone convolutional neural network prototype designed for non-diagnostic oral image classification. We make the following contributions: (1) An ensemble feature-extraction architecture integrating pooled representations from four heterogeneous convolutional backbones—VGG-16 (512-d), ResNet-50 (2,048-d), EfficientNet-B0 (1,280-d), and MobileNetV2 (1,280-d)—producing a unified 5,120-dimensional concatenated latent feature representation classified by a fully connected head with dropout regularization. (2) A decoupled web application serving predictions through an asynchronous FastAPI service with streaming payload boundaries alongside a React single-page interface with client-side canvas blur detection. (3) Runtime uncertainty quantification combining 8-pass Test-Time Augmentation (TTA) and 15-pass Monte Carlo Dropout (MCD) to trigger specialist referral advisories when predictive variance exceeds a calibrated threshold (tau = 0.015). (4) An exhaustive cryptographic SHA-256 hash audit of the repository dataset (940 total JPEG images: 490 cancer, 450 non-cancer across 655 train, 143 validation, and 142 test partitions) that identified 21 exact duplicate image groups encompassing 46 files (25 redundant copies), with 11 groups (25 files, 14 redundant copies) crossing data partitions and compromising test set independence.

Code inspection confirms that previously reported benchmark figures, including an overall 99.92% accuracy, were generated through synthetic random simulation scripts rather than empirical testing on images. True empirical model metrics on real images are currently unavailable. The prototype is explicitly non-diagnostic, lacks regulatory clearance, and is not approved for clinical use.

Keywords: Deep Learning, Oral Cancer Screening, Feature Fusion, Uncertainty Estimation, Monte Carlo Dropout, Medical Image Audit, Web Applications.

I. INTRODUCTION

Oral cancer, primarily manifesting as oral squamous cell carcinoma (OSCC), ranks among the most prevalent malignancies worldwide. According to GLOBOCAN 2020 statistics, cancers of the lip and oral cavity accounted for an estimated 377,713 new cases and 177,757 deaths globally in 2020 [1]. India carries an especially severe disease burden, recording over 135,000 new diagnoses annually—nearly one-third of the global total—driven by cultural and behavioral consumption of chewing tobacco, areca nut, and gutka. Despite surgical and therapeutic innovations, five-year survival rates for advanced OSCC remain below 50%, whereas early-stage detection elevates survival beyond 80%, underscoring the urgent need for accessible screening.

This project was conceptualized and developed as a final-year Bachelor of Engineering capstone project in the Department of Computer Science & Engineering at Kalpataru Institute of Technology, affiliated with Visvesvaraya Technological University (VTU).

Conventional oral examination (COE) through visual inspection and digital palpation under white light is the clinical frontline standard. However, a Cochrane systematic review by Brocklehurst et al. [2] found that while screening targeted high-risk individuals reduces mortality, general population screening suffers from high false-positive rates and inter-examiner variability among general practitioners. Computer vision and deep convolutional neural networks offer potential as exploratory tools to assist frontline personnel in reviewing

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images [3], [4]. However, software tools must be rigorously defined: the prototype presented here is strictly a non-diagnostic decision-support prototype and fundamentally not an autonomous medical device or substitute for biopsy and histopathological verification.

Frontline photographic screening faces several distinct technical challenges:

1) Intraoral Photographic Artifacts: Consumer smartphone cameras capture images under uncontrolled lighting, with severe specular glare from saliva and motion blur degrading classification.

2) High-Dimensional Visual Complexity: Oral mucosal lesions exhibit varied clinical morphologies (erythroplakia, leukoplakia, ulceration) that require diverse feature representations across multiple spatial scales.

3) Overconfident Deep Learning Predictions: Standard softmax classifiers produce high confidence on ambiguous or out-of-distribution inputs, creating safety hazards in healthcare.

4) Methodological Vulnerabilities and Data Leakage: Undetected duplicate images crossing data partitions invalidate reported performance in biomedical machine learning [11], [12].

In this work, we develop and audit MergedNet, a multi-backbone feature fusion prototype for oral image classification. We make the following contributions:

- Multi-Backbone Feature Concatenation: Simultaneous extraction across VGG-16 [5], ResNet-50 [6], EfficientNet-B0 [8], and MobileNetV2 [7] into a unified 5,120-dimensional feature vector.
- Decoupled Enterprise Web Application: An asynchronous FastAPI backend providing ONNX runtime acceleration alongside a React 19 interface.
- Dual-Path Uncertainty Gating: Combining Test-Time Augmentation (TTA) and Monte Carlo Dropout (MCD) to flag ambiguous cases for specialist review.
- Exhaustive Cryptographic Data Audit: Identifying 21 duplicate groups across 940 images and documenting partition crossing.
- Scientific Demarcation: Explicitly separating implemented neural network weights from heuristic rules and disclosing synthetic benchmark origin.

The remainder of this paper is organized as follows. Section II reviews related literature in deep learning oral screening, feature fusion, and biomedical data leakage. Section III details the software engineering architecture, security implementations, and module boundaries. Section IV establishes the mathematical formulations for feature extraction, uncertainty gating, and clinical risk heuristics. Section V presents the experimental testbed, runtime latency benchmarks, cryptographic data audit, and synthetic metric disclosures. Section VI discusses clinical limitations and outlines a six-phase remediation roadmap. Section VII concludes the paper.

II. RELATED WORK

Automated oral cancer screening sits at the intersection of deep learning pathology analysis, computer vision feature fusion, Bayesian uncertainty quantification, and biomedical machine learning governance.

A. Deep Learning in Oral Pathology Screening

Early computational pathology relied on handcrafted textural features such as Gray-Level Co-occurrence Matrices (GLCM) and Gabor wavelets coupled with Support Vector Machines. Welikala et al. [3] demonstrated automated detection of oral lesions using dual-detector ResNet architectures, achieving promising sensitivity but requiring heavily controlled clinical imaging. Warin et al. [4] investigated deep convolutional networks (VGG-19, ResNet-101, and DenseNet-121) across multi-center photographic datasets, demonstrating that network depth enhances diagnostic sensitivity but warning that model generalizability sharply degrades across unfamiliar clinical illumination environments.

B. Multi-Scale Convolutional Feature Fusion

Ensemble feature fusion combines complementary representations from diverse architectures. Simonyan and Zisserman [5] introduced VGG architectures with small 3x3 receptive fields, capturing dense textural patterns. He et al. [6] designed ResNet residual learning frameworks that mitigate vanishing gradients across deep topological representations. Sandler et al. [7] designed MobileNetV2 with inverted residuals and linear bottlenecks for efficient low-latency computation, while Tan and Le [8] developed EfficientNet using compound

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scaling across network depth, width, and resolution. Concatenating intermediate representations across these architectural families allows simultaneous capture of localized textural nuances and high-level morphological structures.

C. Predictive Uncertainty and Test-Time Augmentation

Deep networks trained with standard cross-entropy produce poorly calibrated, overconfident predictions on ambiguous inputs. Gal and Ghahramani [9] developed Monte Carlo Dropout (MCD) as a mathematically grounded Bayesian approximation, sampling multiple stochastic forward passes with active dropout during inference to estimate predictive variance. Ayhan and Berens [10] showed that Test-Time Augmentation (TTA)—generating rotated and scaled input variations—estimates aleatoric uncertainty arising from photographic artifacts, providing an essential safeguard for clinical decision-support.

D. Data Leakage and Deduplication Audits in Medical AI

The biomedical machine learning literature faces severe reproducibility challenges from undetected data leakage. Roberts et al. [11] systematically reviewed 515 published machine learning models for COVID-19 diagnosis, finding that none were clinically viable due to methodological flaws, primarily data duplication and partition leakage. Kapoor and Narayanan [12] demonstrated that data leakage is pervasive across biomedical sciences, artificially inflating test metrics. Rigorous cryptographic hash auditing is therefore a foundational requirement before reporting model metrics.

III. SYSTEM ARCHITECTURE AND PIPELINE

A. System Architecture Overview

The MergedNet platform is engineered as a decoupled, multi-tier web application designed for non-diagnostic clinical decision support. The platform architecture comprises four distinct operational layers:

1) Client-Side Presentation Tier (Frontend): Built with React 19, TypeScript, and Vite, providing a single-page application (SPA) interface. It incorporates a client-side HTML5 Canvas blur pre-check using a Laplacian filter kernel, preventing unsharp uploads before network transmission.

2) API Gateway and Security Tier: Implemented in FastAPI, providing CORS whitelisting, HTTP header security, client-side rate limiting, and request validation through Pydantic schemas.

3) Model Inference and Computational Engine: Houses the MergedNet PyTorch and ONNX neural network inference runtimes alongside non-ML image preprocessing pipelines (Reinhard color transfer, saliva glare inpainting, and OpenCV Laplacian variance calculation).

4) Persistent Data Store: Utilizes SQLite with Write-Ahead Logging (WAL) for persistent session logging, patient record tracking, audit histories, and authentication tokens.

B. Web Application Engineering and Decoupled Architecture

The client application communicates with the FastAPI backend over RESTful HTTP/JSON endpoints. Images captured by mobile or desktop webcams are transmitted as multipart/form-data payloads. The backend validates MIME headers, verifies file size boundaries (<= 10 MB), and checks memory limits before feeding tensors into the inference engine.

C. Security, Hardening, and Concurrency Controls

The web application implements enterprise-grade security controls conforming to OWASP Top 10 guidelines:

- Ingestion Payload Limits: Enforces a strict 10 MB maximum file size limit, reading uploads in 64 KB streaming chunks to prevent memory-exhaustion denial-of-service attacks.
- MIME Whitelisting: Validates file headers against allowed formats (image/jpeg, image/png, image/webp).
- Rate Limiting: Configures SlowAPI rate limiters (5/min for registration, 10/min for authentication, 20/min for prediction endpoints).
- Secure Password Hashing: Implements PBKDF2-SHA256 with 390,000 iterations and 128-bit salt conforming to NIST SP 800-132 recommendations.

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• Singleton Model Concurrency: Protects model forward passes with threading.Lock to serialize GPU/CPU tensor operations.

D. Categorization of Repository Software Modules

To ensure academic precision, Table 3 categorizes all software modules in the repository, explicitly distinguishing neural network code from rule-based and heuristic algorithms. Visual saliency is implemented using Grad-CAM++ [13] on ResNet-50 Layer4 as a mathematical post-processing stage. In addition, the repository file multimodal_fusion.py contains a PyTorch class named MultimodalCrossAttentionFusion; however, live execution tracing reveals that this neural module is completely bypassed during live /predict execution in favor of scripted probability adjustment rules.

Table 3. Categorization of Repository Software Modules (Demarcating Neural Networks vs. Heuristics and Rules)

Functional Component	Code Location	Underlying Implementation Mechanism	Algorithmic Classification
Oral Image Classification	app/services/ml_engine.py	PyTorch MergedNet: 4-backbone feature concatenation [5]-[8]	**Deep Neural Network (Checkpoint Present)**
Image Sharpness Gating	app/services/image_quality.py	OpenCV Laplacian variance calculation (Threshold = 50.0)	**Deterministic Heuristic (Non-ML)**
Saliva Glare Suppression	app/services/optical_normalization.py	Luminance thresholding (B>=238, S<=0.22) + inpainting	**Classical Computer Vision (Non-ML)**
Color Normalization	app/services/optical_normalization.py	Reinhard statistical mean/std transfer in CIE L*a*b* space	**Statistical Color Mapping (Non-ML)**
Lesion Boundary Segmentation	app/services/vision_pipeline.py	Erythema index (R-G)/(R+G) + brightness thresholding	**Handcrafted Heuristic (Non-ML)**
Epidemiological Risk Scoring	app/services/clinical_risk.py	Linear sum: Age (+0.30), Tobacco (+0.35), Betel (+0.30), Alcohol (+0.20)	**Rule-Based Formula (Non-ML)**
Four-Class Ordinal Triage	app/services/multimodal_fusion.py	Scripted probability shifting with bonus weights & temperature scaling (T=2.5)	**Rule-Based Logic (Non-ML)**

IV. MATHEMATICAL FORMULATIONS

A. Optical Normalization and Image Quality Gating

To reduce intraoral illumination variation, input images are normalized in CIE L*a*b* color space using Reinhard statistical color transfer against a calibrated clinical reference distribution:

$$I'_c = (I_c - \mu_c) \times (\sigma'_c / \sigma_c) + \mu'_c \quad (1)$$

where I_c represents channel pixel intensity, μ_c and σ_c are source mean and standard deviation, and μ'_c, σ'_c represent reference distribution statistics.

Image sharpness is evaluated via the variance of the discrete 3×3 Laplacian operator:

$$S_{blur} = \mathrm{Var}(\nabla^2 I_{gray}) \quad (2)$$

Images exhibiting $S_{blur} < 50.0$ trigger an automated low-sharpness warning to the user.

B. Multi-Backbone Feature Concatenation (MergedNet)

MergedNet extracts feature representations across four convolutional backbones with frozen pre-trained ImageNet weights:

• VGG-16: Captures low-level textural patterns through dense 3×3 convolution blocks, pooling into a 512-dimensional vector $f_{vgg} \in \mathbb{R}^{512}$ [5].

• ResNet-50: Captures topological features via identity shortcut connections, producing a 2,048-dimensional vector $f_{resnet} \in \mathbb{R}^{2048}$ [6].

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• EfficientNet-B0: Balances spatial resolution and feature depth through MBConv blocks, outputting a 1,280-dimensional vector $f_{effnet} \in \mathbb{R}^{1280}$ [8].

• MobileNetV2: Provides low-latency depthwise separable convolutions, yielding a 1,280-dimensional vector $f_{mobilenet} \in \mathbb{R}^{1280}$ [7].

The representations are concatenated into a unified 5,120-dimensional latent embedding:

$$f_{merged} = [\, f_{vgg} \,\|\, f_{resnet} \,\|\, f_{effnet} \,\|\, f_{mobilenet} \,] \in \mathbb{R}^{5120} \quad (3)$$

C. Fully Connected Classification Head

The concatenated vector is fed into a classification head with batch normalization and dropout (p = 0.4):

$$z = W_2 \cdot \text{Dropout}_{0.4}(\text{ReLU}(W_1 \cdot f_{merged} + b_1)) + b_2 \quad (4)$$

where $W_1 \in \mathbb{R}^{(512\times5120)}$, $b_1 \in \mathbb{R}^{512}$, $W_2 \in \mathbb{R}^{(C\times512)}$, and $b_2 \in \mathbb{R}^C$ (with class count C = 2 for binary non-cancer vs. cancer classification). Class probabilities are computed via softmax:

$$P(y = c \mid x) = \exp(z_c) / \sum_{j=1}^{C} \exp(z_j) \quad (5)$$

D. Dual-Path Uncertainty Gating and Specialist Referral

To safeguard against overconfident erroneous predictions, the prototype implements dual-path uncertainty quantification combining Test-Time Augmentation (TTA) and Monte Carlo Dropout (MCD):

$$P_{TTA}(y \mid x) = (1 / K) \sum_{k=1}^{K} M(T_k(x)) \quad (6)$$

$$P_{MCD}(y \mid x) = (1 / T) \sum_{t=1}^{T} M(x \mid \hat{W}_t) \quad (7)$$

where K = 8 stochastic spatial transformations are applied for TTA, and T = 15 forward passes are executed with active dropout. The final predictive consensus is:

$$P_{final}(y \mid x) = 0.5 \times P_{TTA}(y \mid x) + 0.5 \times P_{MCD}(y \mid x) \quad (8)$$

Epistemic uncertainty is estimated via maximum class predictive variance across MCD passes:

$$U(x) = \max_c \left[(1 / T) \sum_{t=1}^{T} (P_t(c \mid x) - \bar{P}(c \mid x))^2 \right] \quad (9)$$

When predictive variance exceeds the calibrated threshold $\tau = 0.015$, the prototype overrides autonomous classification:

$$\text{Decision}(x) = \text{'Uncertain - Refer to Specialist'} \quad \text{if} \quad U(x) > 0.015 \quad (10)$$

E. Rule-Based Clinical Risk Scoring

A structured questionnaire collects patient epidemiological risk factors. A non-ML linear formula computes a composite risk score $S_{risk} \in [0, 1]$:

$$S_{risk} = 0.35 \times \mathbb{1}_{tobacco} + 0.30 \times \mathbb{1}_{betel} + 0.30 \times \mathbb{1}_{\{age\geq60\}} + 0.20 \times \mathbb{1}_{alcohol} + 0.25 \times \mathbb{1}_{prior} \quad (11)$$

Scores $S_{risk} \geq 0.60$ trigger an automated clinical alert badge in the user interface.

F. Higher-Order Grad-CAM++ Saliency Formulations

Saliency localization weights w_k^c for convolutional feature map A^k in ResNet-50 Layer4 are formulated via higher-order positive gradients [13]:

$$w_k^c = \sum_{i,j} \alpha_{ij,c}^{k} \times \text{ReLU}\left(\partial^2 Y^c / \partial (A_{ij}^k)^2 \right) \quad (12)$$

Algorithm 1 details the complete end-to-end inference and uncertainty gating procedure implemented in the prototype.

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[t]Algorithm 1 MergedNet Dual-Path Inference and Uncertainty Gating

```
Require: Input image x, model checkpoint M, uncertainty threshold tau = 0.015
Ensure: Class label C, confidence p, uncertainty flag U_flag
S_blur <- Var(Laplacian(Grayscale(x)))
if S_blur < 50.0 then FlagLowQualityAdvisory()
if RuntimeMode == 'PyTorch' then
   EnableDropoutOnly(M)
   for t = 1 to 15 do P_mcd[t] <- Softmax(M(x)) end for
   for k = 1 to 8 do P_tta[k] <- Softmax(M(Transform_k(x))) end for
   P_final <- 0.5 * Mean(P_mcd) + 0.5 * Mean(P_tta)
   U <- Max(Variance(P_mcd))
else if RuntimeMode == 'ONNX' then
   for k = 1 to 3 do P_onnx[k] <- ONNX_Forward(Transform_k(x)) end for
   P_final <- Mean(P_onnx)
   U <- Max(Variance(P_onnx))
end if
if U > tau then
   C <- 'Uncertain (Refer to Specialist)'; U_flag <- True
else
   C <- ArgMax(P_final); U_flag <- False
end if
return {C, Max(P_final), U, U_flag}
```

V. EMPIRICAL AUDIT AND SYSTEM PERFORMANCE

A. Experimental Hardware and Execution Testbed

Software profiling was conducted on an x86-64 server equipped with an Intel Core i7-12700H processor (14 cores, 20 threads at 2.30 GHz), 16 GB DDR4 RAM, and an NVIDIA GeForce RTX 3050 Ti Mobile GPU (4 GB VRAM) running Windows 11. Backend services were executed under Python 3.12 with PyTorch 2.4.0 (CPU execution) and ONNX Runtime 1.19.0. Client-side evaluation was performed across Google Chrome v124 and Mozilla Firefox v125.

B. Dataset Composition and Cryptographic Deduplication Audit

The repository dataset contains exactly 940 oral photographic images partitioned into 490 cancer and 450 non-cancer samples. An exhaustive SHA-256 cryptographic hash audit of all image files revealed significant data redundancy and partition leakage:

- Exact Duplicate Groups: 21 distinct image groups containing 46 total files (25 redundant copies).
- Partition-Crossing Leakage: 11 duplicate groups encompassing 25 files (14 redundant copies) cross data partitions, with identical images present in both training and test partitions.
- Compromised Evaluation Independence: Because 14 test samples have identical duplicates in the training partition, the test set cannot provide independent validation of model generalization.

Table 1. Repository Dataset Partitions and SHA-256 Cryptographic Deduplication Audit

Partition	Cancer	Non-Cancer	Total	Unique SHA-256	Duplicates	Crossing Split
Train	341	314	655	638	17	--
Validation	75	68	143	139	4	4 groups
Test	74	68	142	128	14	11 groups
Audit Groups	--	--	--	--	21 groups (46 files)	11 groups (25 files)
Total	**490**	**450**	**940**	**905**	**25 redundant**	**14 test duplicates**

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C. Inference Latency and Resource Utilization Profiles

Execution latency and memory footprint were benchmarked across PyTorch (CPU) and ONNX Runtime engines over 100 consecutive forward passes, summarized in Table 2.

Table 2. Runtime Latency and Memory Footprint across Execution Engines

Inference Configuration	Execution Engine	Mean Latency	P95 Latency	RAM Footprint
Single Pass (Standard)	PyTorch (CPU)	412.4 ms	468.1 ms	482 MB
TTA Pass (8 variations)	PyTorch (CPU)	3,285.6 ms	3,612.0 ms	510 MB
MCD Pass (15 samples)	PyTorch (CPU)	6,140.2 ms	6,780.5 ms	524 MB
Accelerated Pass (3x TTA)	ONNX Runtime	284.1 ms	318.4 ms	215 MB

D. Repository Audit vs. Synthetic Benchmark Discrepancy

A critical finding of our code audit concerns the origin of previously reported model accuracy. In earlier project documentation, an overall accuracy of 99.92%, precision of 99.90%, sensitivity of 99.95%, and specificity of 99.88% were cited. Code inspection of backend/scripts/ieee_benchmark_evaluator.py reveals that these values were generated via synthetic normal and beta random distributions rather than through empirical model evaluation on image datasets.

Table 4 presents a transparent comparison between the synthetic values and the verified findings from our repository audit.

Table 4. Repository Data Audit vs. Synthetic Simulation Metrics Discrepancy

Metric / Parameter	Reported Synthetic Claim	Audit Finding in Codebase	Scientific Status
Classification Accuracy	99.92%	Scripted random draws (mu=0.9992, sigma=0.0004)	Synthetic Simulation (Unverified)
Sensitivity / Specificity	99.95% / 99.88%	Generated from parameterized Beta distribution	Synthetic Simulation (Unverified)
Dataset Split Independence	Assumed Disjoint	11 duplicate groups (25 files) cross train-test	Partition Leakage Confirmed
Training Pipeline	End-to-end trained	No `train.py` script or training logs in repo	Training History Missing
Uncertainty Gating	Automated Referral	Implemented in code (tau = 0.015, TTA/MCD)	Verified in Codebase

VI. DISCUSSION, CLINICAL IMPACT AND LIMITATIONS

A. Non-Diagnostic Prototype Status and Regulatory Context

A primary imperative of biomedical engineering is the clear distinction between research prototypes and clinical medical devices. MergedNet is an academic prototype designed for experimental decision-support investigation. The system possesses no regulatory clearance (such as US FDA 510(k), European CE mark, or Indian CDSCO approval) and must not be used for autonomous clinical diagnosis. In real-world deployment, all classifications must be supervised by qualified dental and oncological clinicians.

B. Methodological Hazards of Synthetic Evaluation

Reporting model metrics derived from synthetic random generation poses severe ethical and methodological risks in healthcare AI [11], [12]. Synthetic simulations create an illusion of near-perfect accuracy (e.g., 99.92%) that can mislead reviewers and end-users regarding clinical readiness. By openly disclosing that true empirical metrics are currently unavailable pending clean deduplication and retraining, this paper prioritizes scientific transparency over inflated claims.

C. Unverified Dataset Provenance and Ethics Gaps

Our audit identified critical provenance and governance gaps in the local repository:

- Missing Patient Demographics: The repository contains no patient age, sex, lesion anatomical site, tobacco habits, or histopathological stage metadata.

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- Absence of Institutional Review Board (IRB) Documentation: No ethical clearance certificates, informed consent protocols, or clinical governance records are present.
- Unverified Data Source: The dataset is attributed to an unverified secondary Kaggle mirror lacking original clinical trial provenance.

D. Phased Conditional Remediation Roadmap

To provide actionable guidance for future researchers, Table 5 delineates a structured six-phase remediation roadmap required before this prototype can transition toward credible clinical evaluation.

Table 5. Phased Conditional Remediation Roadmap for Clinical and Academic Translation

Phase	Remediation Focus	Key Engineering & Methodological Objectives
Phase 1 (Conditional)	Provenance & Governance Verification	Identify original Kaggle dataset source URL; verify licensing, patient consent, and ethics permissions.
Phase 2 (Conditional)	Deduplication & Patient Partitioning	Purge 21 exact duplicate image groups; establish strict patient-level separation into disjoint splits.
Phase 3	Reproducible Training Pipeline	Develop version-controlled `train.py`; log optimizer, learning rate schedule, loss criteria, and seeds.
Phase 4	Empirical Evaluation & Ablation	Evaluate retrained MergedNet on independent test split; compute true empirical metrics (Accuracy, F1, AUROC).
Phase 5 (Conditional)	Auxiliary Module Refinement	Replace heuristic segmentation with validated models; calibrate risk formulas against clinical literature.
Phase 6 (Conditional)	Clinical Partnership & Reader Trials	Upon formal ethical approval, collaborate with oral oncologists to conduct blinded reader evaluation.

VII. CONCLUSION AND FUTURE WORK

This paper has presented the software engineering design, inference pipeline, and repository data audit of MergedNet, a multi-backbone feature fusion prototype for non-diagnostic oral image classification. The system extracts and concatenates visual embeddings from VGG-16 [5], ResNet-50 [6], EfficientNet-B0 [8], and MobileNetV2 [7] into a 5,120-dimensional latent representation, served through an asynchronous FastAPI service and a React frontend with runtime uncertainty gating logic [9], [10].

Rather than presenting unverified metrics, this paper provides an honest, evidence-based account: the prototype is non-diagnostic; the dataset contains 940 images with 21 duplicate groups (11 crossing splits) and unverified provenance; training scripts are missing; previously reported 99.92% accuracy was generated through synthetic random simulation; and true empirical metrics on real images are currently unavailable. Prioritizing scientific integrity, data auditing, and transparent systems documentation provides a credible baseline for future academic research.

The authors express their sincere gratitude to Project Guide Prof. Aishwarya S, Assistant Professor, Department of Computer Science & Engineering, Kalpataru Institute of Technology, Tiptur, for technical guidance and supervision. The authors also acknowledge the Department of Computer Science & Engineering, Kalpataru Institute of Technology, and Visvesvaraya Technological University (VTU), Belagavi, for providing laboratory facilities and institutional support.

We intend to explore future work across several technical avenues:

- Multi-Class Ordinal Classification: Transitioning from binary classification to a 4-class ordinal model (Healthy -> Benign -> OPMD -> Malignant) using ordinal cross-entropy loss.
- Hybrid Vision Transformers: Integrating a Swin Transformer branch to capture long-range tissue structural atypia alongside local CNN features.
- Calibrated Uncertainty: Implementing temperature scaling and conformal prediction for rigorously calibrated decision-support referral boundaries.

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